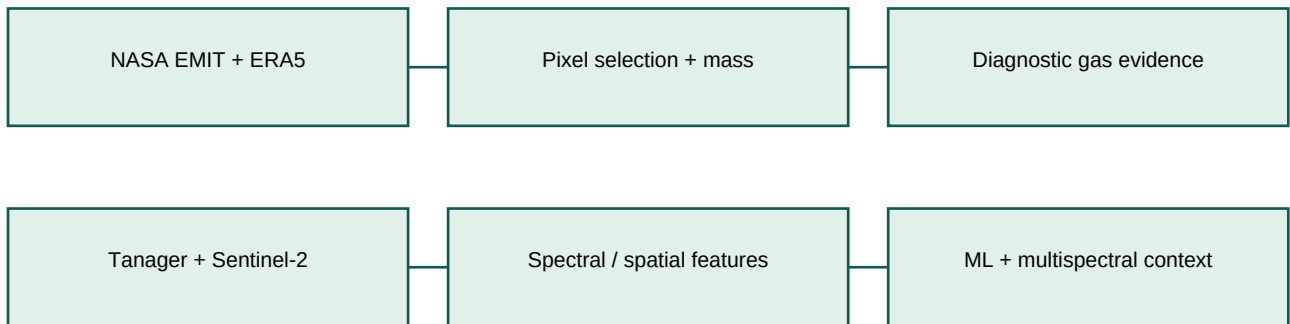


Methane and carbon intelligence

Challenge: Air Intelligence | Team: EcoOrbit

Solution and evidence boundary

A reproducible proof of concept for reviewing real methane plume pixels, wind, selected excess mass and diagnostic flux. Published CO2 observations provide separate reference evidence; EcoOrbit has not retrieved landfill CO2 pixels. NO2/SO2 and multispectral site context support follow-up.



Real observed cases

Event	Peak ppm m	Wind m/s	Selected excess CH4 kg
jeddah 20240609	4476	1.14	2220
seropedica 20240305	11076	1.94	6413
seropedica 20240924	2511	2.73	1399

Mass is a selected snapshot quantity, not an emission rate. Jeddah uses source-scene sensitivity and uncertainty; Brazil uses original plume rasters without inventing companion corrections. Flux scenarios are unvalidated and excluded from annual inventories.

Preprocessing implemented in code

Stage	Provider input / EcoOrbit operation	Code
Acquire	Real EMIT TIFF/GeoJSON and ERA5; Tanager source scene IDs and site registry.	emit_cases.py; multisite_download.py
Calibration / atmosphere	Provider gas retrieval and L2 surface products. EcoOrbit does not rerun radiance calibration or atmospheric correction. Landsat values are scaled to Celsius.	site_heat.py; train_classifier.py
Geometry / QA	Coordinate transforms, local sampling, cloud/cirrus/no-data rejection; actual WGS84 plume pixel areas.	train_classifier.py; emit_pixels.py
Spectral binning	11 x 11 Tanager site windows, ≥ 50 valid pixels; 30 nm bins excluding strong water absorption; spatial medians and indices.	train_classifier.py
Methane preparation	Jeddah: ENH/SENS, background subtraction, sensitivity 0.5-2 and SNR ≥ 2 . Brazil: vetted PLM mask and enhancement thresholds only.	emit_pixels.py; emit_cases.py
Multispectral preparation	Sentinel-2 B04/B08/B11 + SCL; 20 m grid, site/reference NDVI/NDBI. Landsat QA and site/reference surface temperature.	site_land.py; site_heat.py

Analysis and machine learning

Gas analysis: convert ppm m to column mass using the ideal gas law and pixel area; sum selected excess methane. Review thresholds of 500, 1000 and 1500 ppm m. ERA5 supplies real hourly overpass context. Diagnostic $Q = 3600 \times \text{wind} \times \text{integrated mass} / \text{maximum selected-pixel separation}$.

This is processing of NASA spectral retrievals, not a new matched-filter retrieval from radiance. Whole-complex geometry, coarse wind, unverified source origin and incomplete uncertainty prevent a validated facility rate.

Supplementary ML: median imputation, standardization, PCA and balanced logistic regression. A fixed city/label-stratified split uses 36 training and 12 test sites across all three cities. Three-fold training-only validation chooses 4/8/12 components. Observed accuracy is 11/12 (91.7%), AUC 0.944.

Labels distinguish industrial registry sites from school controls. This small screen does not establish universal factory/landfill accuracy and does not estimate methane. Green image patches provide separate vegetation context.

Multispectral example

Sentinel-2 L2A provides two-date vegetation and built-surface indices; Landsat C2 L2 provides surface-temperature contrasts. Completed US observations are separate in location and date from the landfill events. They demonstrate additional site investigation, not plume attribution.

Carbon and reporting

A source-reported CO₂ reference at an Indian power-generation site demonstrates published hyperspectral carbon evidence. It is not a new EcoOrbit retrieval. GRI 305 and GHG Protocol evidence mapping distinguishes satellite observations from activity-based annual inventories; no annual Scope 1/2 CO₂e is inferred.

Judge walkthrough and reproduction

Solution: select one of three real methane observations, inspect pixels and wind, compare mask thresholds and diagnostic flux, and inspect separate published CH₄/CO₂ references.

Methodology: review provider preprocessing, actual code operations, ML evaluation and reporting boundaries.

Additional features: inspect US site maps, Tanager material/site classification, Sentinel-2 changes, Landsat heat and relevant NO₂/SO₂.

From the project root:

```
python -m pip install -r requirements.txt
python -m src.emit_cases
python -m src.evaluate_sites
python -m src.build_website
```

Use notebooks/02_main_analysis.ipynb and notebooks/04_emit_pixel_analysis.ipynb. Real processed inputs and small source crops are bundled; full raw imagery has documented download commands. Colab downloads a standalone HTML website without a localhost proxy.

Primary methods and provenance

NASA EMIT trace-gas ATBD and user guide:

<https://github.com/emit-sds/emit-sds-tgp/tree/develop/docs>

ERA5 via Open-Meteo historical archive:

<https://open-meteo.com/en/docs/historical-weather-api>

Exact inputs, hashes and processing limitations: results/real/emit_pixels/case_summaries.json and docs/EMIT_PIXEL_METHOD.md.

Detailed gas thresholds and source-scene IDs are retained in the input/output manifests. Only real observations are used in this release.